

CONTRIBUTING TO FINOMOP MAPPINGS

A clinician friendly guide

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- This guide is designed to be as accessible as possible, not requiring previous experience of git or version control, Usagi, and so on. However, basic knowledge of OMOP is required. Things like what is a vocabulary will not be discussed here.

THIS GUIDE IS ARRANGED AS FOLLOWS:

- 1) Set up **Usagi** and the required files.
- 2) Do changes to mappings / make new mappings
- 3) Basic use of github to add your contributions

CONTRIBUTING TO FINOMOP MAPPINGS

This doc is only a starter. Further, more advanced topics are covered in the repos documentation section. See, for example. [FinOMOP mappings/DOCUMENTATION](#)

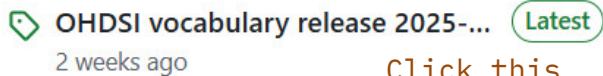
First you will need to have access to the vocabulary files: instructions of getting access to the FinOMOP git, where the vocabularies reside are in the [public landing_page for FinOMOP](#)

FinOMOP mappings use the usagi file format. Next, we will see how to get and use Usagi.

USAGI SETUP 1: Downloads

- You need: **OHDSI vocabularies** from the FINOMOP OMOP-Vocabularies repo 2) **USAGI** and **Java** (see end of slide)
- The vocabularies you need are in the /releases. Download the zip file

Releases 3



+ 2 releases



Download the latest Usagi jar file <https://github.com/OHDSI/Usagi/releases/>
Usagi is a OHDSI tool, it has nothing to do with FinOMOP.

Usagi requires Java. If you dont have it, for ex., from <https://openjdk.org/> following the directions there. FINOMOP has nothing to do with Java.

USAGI SETUP 2:

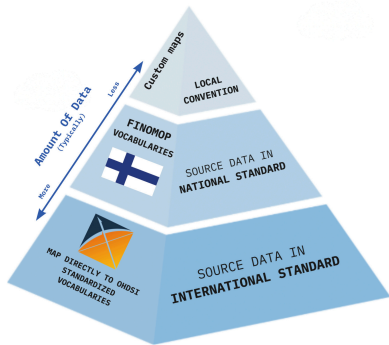
Launching Usagi and loading vocabularies

Launching Usagi

- Usagi is a .jar file. You open it with Java.
- For example on Windows, if you downloaded and installed OpenJDK you right-click on the Usagi file, choose Open with, and point to where java.exe is. Typically it is in a folder under Program Files like `C:\Program Files\Java\jdk-23.0.1\bin`. For Linux or further support see for ex., <https://askubuntu.com/questions/322458/how-to-open-jar-file-using-open-jdk>
- The author starts Usagi from windows commandline by calling `C:\Users\XXXX\Documents\USAGI> java -jar .\Usagi_v1.4.3.jar`

WHY USAGI?

- Partly due to history: the FinOMOP vocabulary files have been created in usagi and the file format has stuck.
- Using Usagi also makes the user to convey to a format common for all mappers.



The picture on the left, from the vocabulary page in yhteistyötilat explains how FinOMOP vocabularies relate to OHDSI vocabularies and source data.

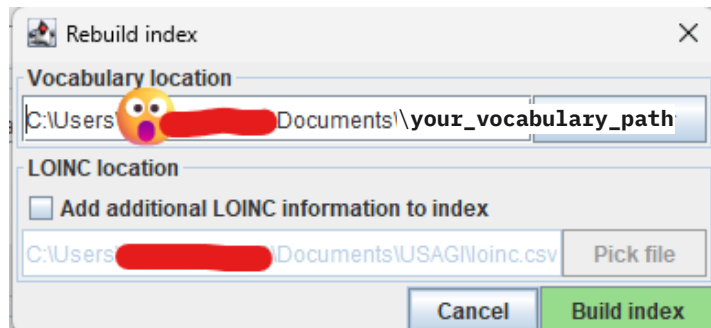
The mappings we make replace or augment mappings of OHDSI for finnish, national codes.

These replacing concepts and alterations are what the usagi files hold.

USAGI SETUP 2:

Loading vocabularies

- Once you open Usagi, it will kindly ask you for the location of the vocabularies you want to load.
- Point this to where you saved and unzipped the vocabularies. (input_omop_vocabulary file)



Building index. This will take a while....
Loading parent-child information

Wait for the index to build.

After you've loaded the vocabularies you can use USAGI to make your own, new or proposed mappings. If this was your goal, then you can stop reading this tutorial here!
If you are adding a completely new vocabulary see below link.

https://github.com/FinOMOP/FinOMOP_mappings/blob/main/DOCUMENTATION/how_to_add_a_new_vocabulary.md

USAGI SETUP 2:

Loading vocabularies: common questions and problems

In this tutorial, we used the FinOMOP mapping files, because we want to change them or update them.

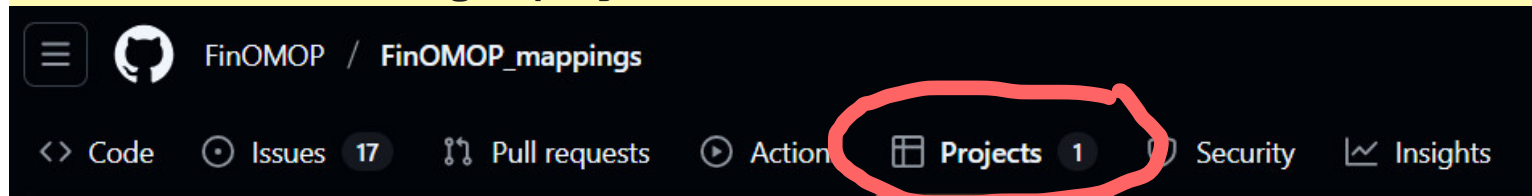
If, for example, you would be creating new mappings, you could very well just get the **target vocabularies from Athena**.

In this case if you would know which vocabulary it is you want to map to you could just download a **subset of the vocabularies**

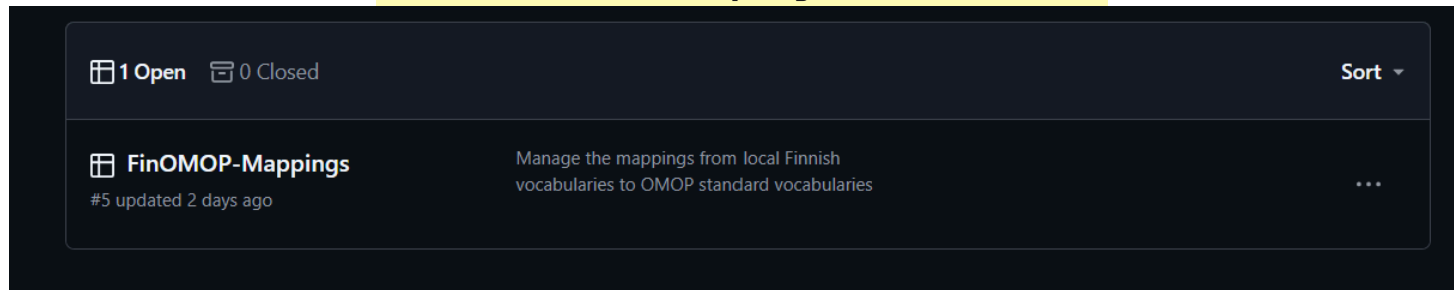
This makes the index building time faster.

USE CASE: CONTRIBUTING TO FINOMOP MAPPING / VOCABULARIES

Here we depict the process through a real-world example of solving an issue on the FinOMOP git project



You can find the project board here

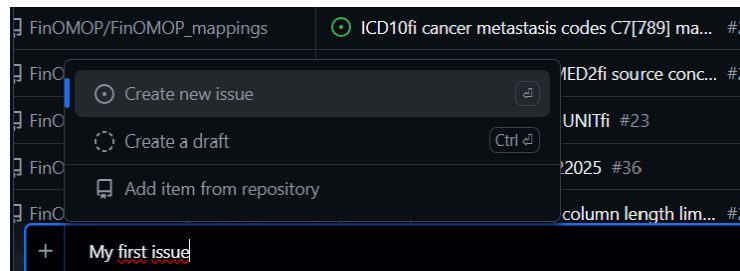


Choose the FinOMOP mappings board

	Repository	Title	Release	Status	Assignees	Labels	Priority
1	FinOMOP/FinOMOP_mappings	Add Drug route vocabulary #25	No DL	Backlog	ahammais		
2	FinOMOP/FinOMOP_mappings	Add drug-related units from HUS drug data ... #64		Backlog	ahammais		
3	FinOMOP/FinOMOP_mappings	Issue withAA1AA (NCSPfi) #62	Monthly (DL 2 Jul.)	Backlog	vajnie	good first issue	
4	FinOMOP/FinOMOP_mappings	Harmonise finnish unit values #39		Backlog			
5	FinOMOP/FinOMOP_mappings	Create templates #42		Backlog	shanjumukhasampath	documentation	

In FinOMOP we track jobs through git issues.

- Above, you can see the issue listing and who it is assigned to.
- Here, we will look at the Issue "Issue withAA1AA (NCSPfi) #62" assigned to the author
- If you want to make a change or fix something in the mappings, first an issue like this should be created with a description of what needs to be done.
- Creating an issue would work so, that you



On the left you can see that you can create an issue in the bottom of the page.

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Here is a well defined and explained issue



javier-gracia-tabuenca-tuni opened 2 weeks ago

Source code: AA1AA

Source vocabulary: NCSPfi

Source concept name: CT of head and brain

Source concept id: 2001300105

Status: MAPS TO 1

Target concept ids: 4125350

Target concept names: CT of head

Target vocabulary: SNOMED

Please describe the issue hee:

Name should be: Head X-ray examination

source : <https://www.terveysportti.fi/terveysportti/toimenpideluokitus.koti>

Create sub-issue

First thing we do, is create a branch where we are going to solve this.

This is essentially a copy of the repository development branch where we can work without messing others things up.

When we are ready, ask for review and make a pull request to admin to *merge* our work to the development branch (and later to main branch).

Development



[Create a branch](#) for this issue or link a pull request.

Now, you still need to switch to that branch before making changes!

This happens on the repository side!

FinOMOP_mappings Public

main 9 Branches 0 Tags

Go to file Add file Code

Switch branches/tags

Find or create a branch...

Branches Tags

- ✓ main default
- 56-update-finngen-code-counts-to-df13
- 62-issue-withaa1aa-ncspfi
- 64-add-drug-related-units-from-hus-drug...
- 67-adding-code-combinations-to-icd10fi
- development
- gh-pages
- update-dahsboard
- update-documentation

View all branches

CONVENTIONS.md 7fc4c61 · 2 days ago 136 Commits

- aaa 3 weeks ago
- updated 2 weeks ago
- update documentation 3 weeks ago
- Create CONVENTIONS.md 2 days ago
- updated dashboard to include issue report link in tables 2 weeks ago
- gitignore .DS_Store 2 months ago
- Initial commit, copied from FinOMOP v2.1.0 release 3 months ago
- Enhance README.md with detailed repository information, i... 3 weeks ago

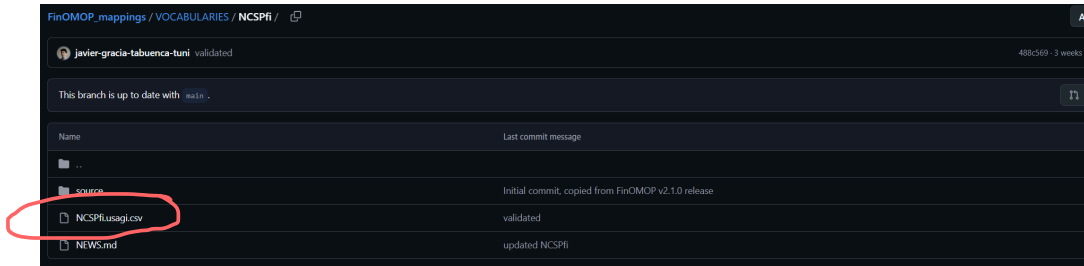
For more information how the repositories are organized and named see [repo rules and nomenclature](#)

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Now, we make our changes to the usagi file, that acts as a basis of our mappings.

Download the specific usagi file.

Our task here was to change the NSCPfi mapping concept with Source code **AA1AA** from its wrong name CT of head and brain to "Head X-ray examination"



Now, you fire up usagi, with

1. The Athena vocabs from OMOP_Vocabularies repo
2. The file you want to modify : from Usagi, you select File > import codes

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Usagi - NCSFpi.usagi.csv

File Edit View Help

Status	Source code	Source term	Frequency	Match score	Concept ID	Concept name	Domain
--------	-------------	-------------	-----------	-------------	------------	--------------	--------

Import codes from NCSFpi.usagi.csv

sourceCode	sourceName	sourceFrequency	matchScore	mapping	equivalence	statusSetBy	statusSet	conceptId	conceptName	domainId
A	Nervous ...	-1	0	APPROVED UNREVIE...	kalleasep...	15909696...	4096246	Structure ...	Spec Ana...	
AA1A	Head X-r...	1744	0	FLAGGED UNREVIE...	tays	15961536...	4184771	Plain X-ra...	Procedure	
AA1AA	CT of hea...	61318	0	APPROVED UNREVIE...	tays	15961536...	4125350	CT of head	Procedure	
AA1AG	Short MRI...	1132	0	APPROVED UNREVIE...	tays	15961536...	4082979	MRI of head	Procedure	
AA1AM	Short MRI...	203	0	APPROVED UNREVIE...	tays	16250112...	37311324	MRI of brain	Procedure	
AA1BA	Extensive...	42	0	APPROVED UNREVIE...	Automatic...	17476375...	4155446	Plain X-ra...	Procedure	
AA1BD	Extensive...	7215	0	APPROVED UNREVIE...	tays	15961536...	4125350	CT of head	Procedure	
AA1BF	Standard ...	14	0	APPROVED UNREVIE...	TYKS	16243200...	4082979	MRI of head	Procedure	
AA1BG	Standard ...	9785	0	APPROVED UNREVIE...	tays	15961536...	4082979	MRI of head	Procedure	
AA1BM	MRI exam...	1117	0	APPROVED UNREVIE...	tays	15961536...	4082979	MRI of head	Procedure	
AA1CD	Very exte...	6245	0	APPROVED UNREVIE...	tays	15961536...	4125350	CT of head	Procedure	
AA1CF	Extensive...	-1	0	APPROVED UNREVIE...	TYKS	16243200...	4082979	MRI of head	Procedure	
AA1CG	Extensive...	18020	0	APPROVED UNREVIE...	tays	15961536...	4082979	MRI of head	Procedure	
AA1CM	Extensive...	6099	0	APPROVED UNREVIE...	tays	15961536...	4082979	MRI of head	Procedure	
AA1DD	Head and...	95	0	APPROVED UNREVIE...	tays	16250112...	4324344	CT of hea...	Procedure	
AA1DG	Very exte...	7075	0	APPROVED UNREVIE...	tays	15961536...	4082979	MRI of head	Procedure	
AA1DM	Very exte...	1482	0	APPROVED UNREVIE...	tays	15961536...	4082979	MRI of head	Procedure	
AA1FD	Trauma C...	-1	0	APPROVED UNREVIE...	TYKS	16243200...	4075967	Examinati...	Procedure	
AA1FD	Trauma C...	-1	0	APPROVED UNREVIE...	TYKS	16243200...	4254070	CT of enti...	Procedure	
AA1XD	Other spe...	-1	0	APPROVED UNREVIE...	TYKS	16243200...	4254070	CT of enti...	Procedure	
AA3AA	Facial bo...	424	0	APPROVED UNREVIE...	Automatic...	17476375...	1075146	Plain X-ra...	Procedure	
AA3AD	CT of face	663	0	APPROVED UNREVIE...	tays	16239744...	4061008	CT of face	Procedure	
AA3AI	Cone bea...	1568	0	APPROVED UNREVIE...	tays	15961536...	37397604	Cone bea...	Procedure	

Column mapping

Source code column

Source name column

Source frequency column

Auto concept ID column

Additional info column

sourceCode

sourceName

sourceFrequency

In the dialogue that opens you have to specify a few things:

You only need to insert the first three, they should always have the same column names.

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Select the concept

CT of head and brain	61318	0.00	0	Unmapp...
Short MRI examination of brain with low intensity magnet		0.00	0	Unmapp...
Short MRI examination of brain with low intensity magnet	4400	0.00	0	Unmapp...

Make your change

Source code	Source term
	CT of head and brain



Source code	Source term
AA1AA	Head X-ray examination

Save the file!

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

ONE MORE THING!

FinOMOP_mappings / VOCABULARIES / NCSPI /

javier-gracia-tabuenca-tuni validated 488c569 · 3 weeks ago History

Name	Last commit message	Last commit date
..		
source	Initial commit, copied from FinOMOP v2.1.0 release	3 months ago
NCSPI.usagi.csv	validated	3 weeks ago
NEWS.md	updated NCSPI	3 months ago

vajnir Fixed issue 62 eda1928 · now History

Preview Code Blame 35 lines (21 loc) · 1.19 KB

Raw Download Edit

NCSPI 1.1.6

Fixed 62-issue-withaa1aa-ncspfi : change the NCSPI mapping concept with Source code AA1AA from its wrong name CT of head and brain to "Head X-ray examination"

NCSPI 1.1.5

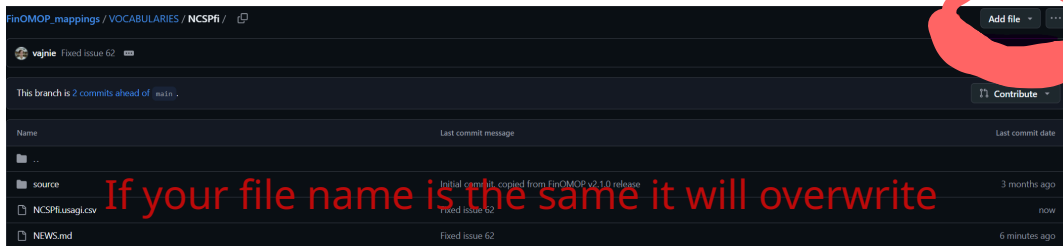
- Fixed 2 domain combination errors after update

At FinOMOP repository inform others of the changes youve made!

Open NEWS.md file and write down the changes you made.

In most cases, you can only use the third number (minor fix).

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING



After you are done with your changes:

1. Upload (replace) changed file to the branch you created

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

FINALLY:

Create a pull request on your branch. The pull request should point to a merge with the **development branch**



What this does, is:

- 1) Runs test and informs you of errors
- 2) Sends a message for a maintainer to check your work.

After its accepted, it will be included in the repository (merged) and then, when a new release comes, be a part of that release.

Congratulations! You are now a FinOMOP contributor!